

When gold moves but trade does not: nonmonetary gold in the U.S. trade statistics

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Between December 2024 and March 2025 the United States imported roughly 870 tonnes of gold more than its own decade-long trend implies — about a quarter of annual world mine production — and by mid-2026 most of it had gone back out again. This letter shows that the episode was a change of address rather than a change in trade. The metal moved when, and only when, the price of gold *in New York* rose far enough above the price of gold *in London* to cover the cost of flying it across the Atlantic. That cost turns out to be very small: the whole round trip consumed an estimated \$243 million of real resources, 0.14% of the \$174 billion of metal it shuttled. The measurement consequences were not small. Gold accounts for 26% of the record January 2025 trade deficit and 41% of the unusually narrow October 2025 figure, and the twelve-month deficit was overstated by \$79 billion at its peak. The national accounts already exclude nonmonetary gold for precisely this reason. The trade release does not, and we argue it should.

Gold is an awkward object for a trade statistician. Most of what crosses a border does so because somebody intends to consume it, install it, or turn it into something else. Gold bullion frequently crosses a border for none of those reasons. It moves because its owner wants it held somewhere else — in a different vault, under a different legal wrapper, in a different jurisdiction — while remaining, economically, the same asset belonging to the same person. Nothing has been produced and nothing has been consumed. An address has changed.

Customs authorities record the crossing all the same, because a crossing is what they are built to observe. In ordinary times this matters little: monthly U.S. imports of nonmonetary gold ran between one and four billion dollars for most of the past decade, a rounding error against goods imports of roughly \$270 billion a month. Between December 2024 and March 2025 they did not. Imports reached \$34.2 billion in January 2025 alone,¹ and the three largest monthly U.S. trade deficits in the history of the series — which begins in 1992 — fall consecutively in that window.

This letter asks what happened, what it cost, and what it did to the published numbers. We take no position on the trade policy that set it off. The trigger is treated here purely as a date after which the expected tariff treatment of gold imports became uncertain, and the question is what the statistical system did with the flow that followed.

Gold behaved like nothing else in the trade data

Figure 1 places gold among the goods the United States actually trades. Each point is one product or one partner, positioned by average monthly exports and imports on logarithmic axes, in three phases: a baseline year, the five-month surge, and the reversal that followed. Above the diagonal the United States is a net importer of that item; below it, a net exporter. A commodity whose trade grows normally drifts outward along the diagonal. A commodity the United States starts buying moves up.

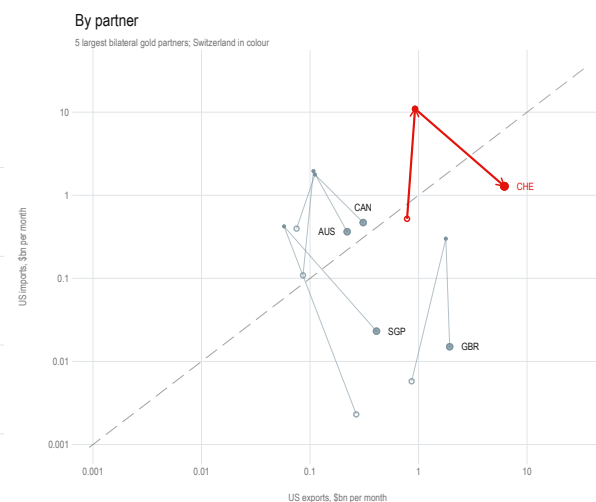
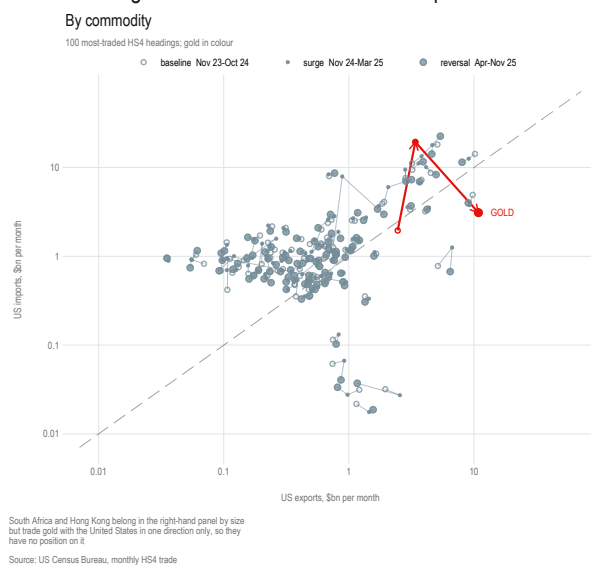
Gold does something else entirely. It moves sharply up and to the left into the surge — imports multiplying while exports fall — and then travels back down and to the right, ending the reversal phase further from the diagonal on the export side than where it began. Switzerland, in the right-hand panel, traces the same shape for the same reason: it is where the bars are recast into the sizes the New York exchange accepts.² Of the 100 most-traded headings, no other traces an out-and-back path of remotely this size.

That shape is the first piece of evidence, and it is worth being precise about why. Metal that has been absorbed — fabricated into jewellery, bought as a retail coin, consumed in electronics — does not come back. Metal that has been relocated does, as soon as the reason for relocating it disappears. A round trip in the trade data is the signature of relocation, and it is not a signature that ordinary imports produce.



Gold went in, then came back out

US trade on the balance plane, monthly average within each phase.
Above the diagonal the United States is a net importer



1. U.S. trade on the balance plane, by commodity and partner, November 2023–November 2025

Notes: Each point is the monthly average within a phase, so that phases of different length are comparable. Both axes are logarithmic, so a series must trade in both directions in all three phases to appear. Gold combines HS 7108 and 7115.

Sources: U.S. Census Bureau.

The price of location, not the price of gold

If the metal moved for a reason, the reason should be visible in a price. It is, but not in the price of gold. It is in the difference between the price of gold in two places.

The relevant object is familiar to any international economist under a different name. Covered interest parity says that the forward premium on a currency must equal the interest differential, or a riskless arbitrage exists. Gold obeys the same logic with the interest rate replaced by the cost of storing metal:

$$F(\tau) = S \cdot e^{b\tau},$$

where S is the spot price, $F(\tau)$ the futures price for delivery τ days out, and b the cost of carrying an ounce for a day — financing, storage and insurance, net of any income from lending the metal out. If the futures curve satisfies this relation, a gold futures contract is a claim on the same metal at a later date, and nothing more.

It need not. Deliverable futures are claims on metal *in a particular place*. A COMEX contract delivers in New York; the London benchmark prices metal in London. When the two diverge by more than carry, the gap is not compensation for time. It is the price of location.

We estimate that price directly. Each trading day we project the whole COMEX futures curve onto a line in log price against days to delivery, weighting each contract by its open interest:

$$\ln F_i(\tau_i) = a + b \tau_i + u_i.$$

The slope b is the carry. The intercept a is what the curve implies New York gold is worth for immediate delivery, and the premium is that intercept over the London price, $p = a - \ln S$. This is a reduced form: p is defined by the projection, and we make no attempt to split it into its location, form and trust components, which we do not believe are separately identified.

One correction matters more than it sounds. The COMEX settlement is struck at 13:30 in New York and the London benchmark at 15:00 in London — three and a half hours apart, or two and a half for the three weeks each year when one country has changed its clocks and the other has not. Gold moves in that gap, and the movement enters p as pure measurement error. Because it is common to every contract on a given day, it lands entirely on the intercept and not at all on the slope, so it can be removed by pricing both legs at the same instant from minute data. Doing so cuts the standard deviation of the daily premium from 0.499% to 0.235%. More than half the apparent day-to-day variation in the spread was an artifact of comparing two prices struck at different times.

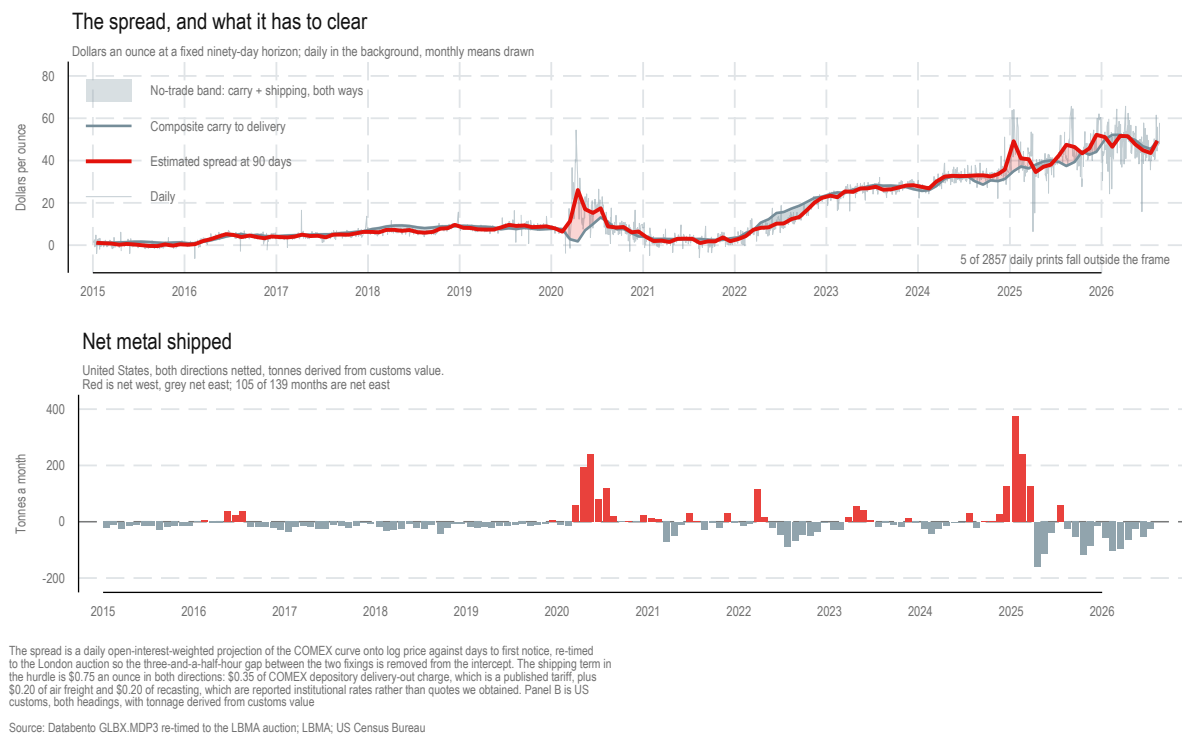
Figure 2 plots the result against what an arbitrageur must clear: the carry to delivery plus the one-off cost of flying and recasting an ounce. Below that band, moving metal loses money and nothing should move. Above it, the trade pays.

The lower panel shows what did move. Over 2,862 trading days the spread cleared the westward threshold on 835 and the eastward one on 1,169 — normally the United States is a net *exporter* of gold, and for most of the sample the spread sits where that is the profitable direction. In the episode both series move together: the premium rises, and 867 tonnes go west on net between December 2024 and March 2025. From April 2025, once gold was excluded from the announced tariff schedule and the premium fell back through the band, 281 tonnes come back.

The relationship is not mechanical, and the figure is honest about that. The spread cleared the cost of shipping in many months when very little metal moved. Clearing the threshold makes a shipment profitable; it does not compel one. What the figure establishes is the weaker but sufficient claim: the metal moved when the price of location made it worth moving, and in the direction that price pointed.

Gold moves only when the spread clears the cost

The estimated COMEX-London spread against carry plus shipping, and the net metal that crossed. Above the band the westward trade pays; below it the eastward one does



2. The COMEX–London spread against the cost of moving metal, and the net flow, 2015–2026

Notes: The spread is a daily open-interest-weighted projection of the COMEX curve onto log price against days to first notice, re-timed to the London auction. The shipping component of the hurdle is \$0.75 an ounce in both directions, of which \$0.35 is the published COMEX depository delivery-out charge. Tonnage in the lower panel is derived by dividing customs value by the London benchmark.

Sources: Databento (CME Globex MDP 3.0); LBMA; CME Group; U.S. Census Bureau.

About 870 tonnes crossed that otherwise would not have

How much of the flow was the episode, and how much was the trade that would have happened anyway? Figure 3 answers this in the simplest available way. We fit a straight line by least squares to all 118 months of U.S. gold imports from January 2015 to October 2024, carry it forward unchanged, and measure the area between the actual series and that line.

The pre-period is deliberately the whole of it rather than a recent slice. A short window invites the objection that it was chosen to flatter the result; a decade does not, and the fitted line is drawn across the entire chart so the reader can judge the fit against the data that produced it. The line rises at 0.090 tonnes a month — U.S. gold imports were very gently trending up — and reaches 41 tonnes by January 2025 against an actual figure nine times larger.

The five-month episode runs **873 tonnes** above the line. Over the whole 21 months since the break, and netting off the fifteen months that fall *below* it, the excess is 540 tonnes. For scale, the episode figure is about a quarter of annual world mine production. It is a great deal of metal to move for a reason unrelated to anybody wanting to own more of it.

Two caveats belong with that number. It is a difference from an extrapolated line, not a causal estimate: anything else moving gold westward in those months is counted here as episode excess. And because U.S. customs data report value rather than weight for these headings, the tonnage is derived by dividing by the London benchmark, which makes it a slightly different measurement from the Swiss and British export figures that carry mass directly.

Moving it was remarkably cheap

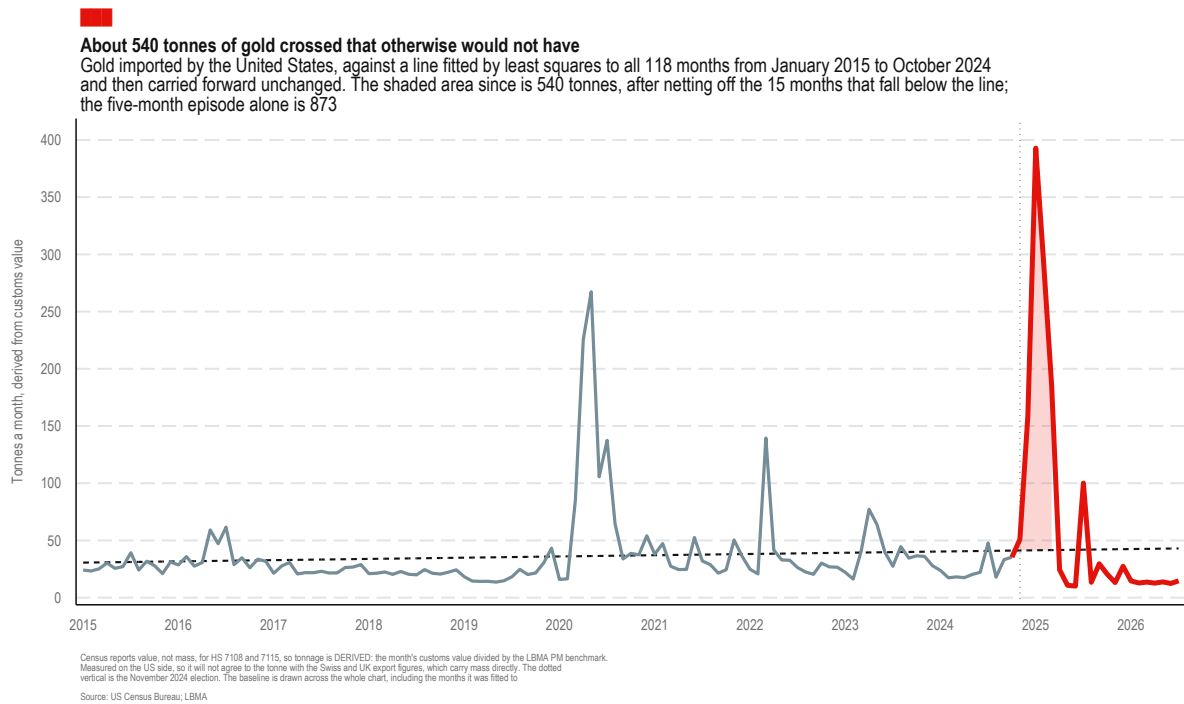
What did the relocation consume? The question is worth asking precisely, because it is the one place where a reassuring answer is the correct one.

Freight is charged by weight, recasting by the bar, and exchange handling by the ounce, so those three are fixed dollar amounts that do not move with the gold price. Insurance and the financing of metal in transit are charged on value, so they roughly doubled over this sample because gold did. Adding them gives an all-in cost of \$1.70 to \$10.72 an ounce for one leg, with a central estimate near \$4.94.³

Applying that to the excess metal in both directions — 1,632 tonne-legs, since a round trip is two flights and two recastings — gives a bill of **\$243 million**, with a range of \$85 million to \$520 million. Set against the \$174 billion of metal those legs carried, that is **0.14% of the value moved**.

This is the central fact about the episode's real consequences, and it cuts in an unglamorous direction. Relocating gold is close to free relative to what gold is worth. That is precisely why a few dollars an ounce of price differential was able to move a quarter of a year's mine output in five months: the barrier to moving it was never high. It also means the episode destroyed very little. No factory was idled and no consumer went without. Roughly a quarter of a billion dollars of freight, insurance, refining and financing was consumed to move metal into a vault it subsequently left — wasteful in the narrow sense, and trivial in any aggregate sense.

The consequences that are *not* trivial are informational.



3. U.S. gold imports against a pre-episode trend, 2015–2026

Notes: The baseline is a least-squares line fitted to all 118 months before November 2024 and carried forward unchanged; it is not clipped, and it is drawn across the months it was fitted to as well as the months it projects. Tonnage is derived from customs value at the London benchmark.

Sources: U.S. Census Bureau; LBMA.

What it did to the published trade figures

Gold reaches the trade release twice, by two independent routes. The first is the nominal headline. The second, less visible and harder for a reader to undo, is the price index used to deflate it.

First, the nominal headline

Figure 4 puts the monthly U.S. goods-and-services deficit next to the same balance with nonmonetary gold removed from both sides. The adjustment is one line of arithmetic: the published balance is exports minus imports, so taking gold out of both gives the balance plus net gold imports. We subtract it *from* the published series rather than rebuilding a deficit from scratch, so the two lines differ by the gold and by nothing else.

The gap is large exactly when the headline was most discussed. January 2025 printed a record \$124.7 billion deficit; without the gold it is \$92.2 billion, so **26% of the record was bullion**. October 2025 printed \$37.4 billion, the narrowest in six years and widely read as an improvement in the trade position; without the gold it is \$52.6 billion, so **41% of the improvement was bullion leaving the country**. The same mechanism produced three consecutive record deficits and, seven months later, the best trade month in six years. Neither was trade in any economically meaningful sense.

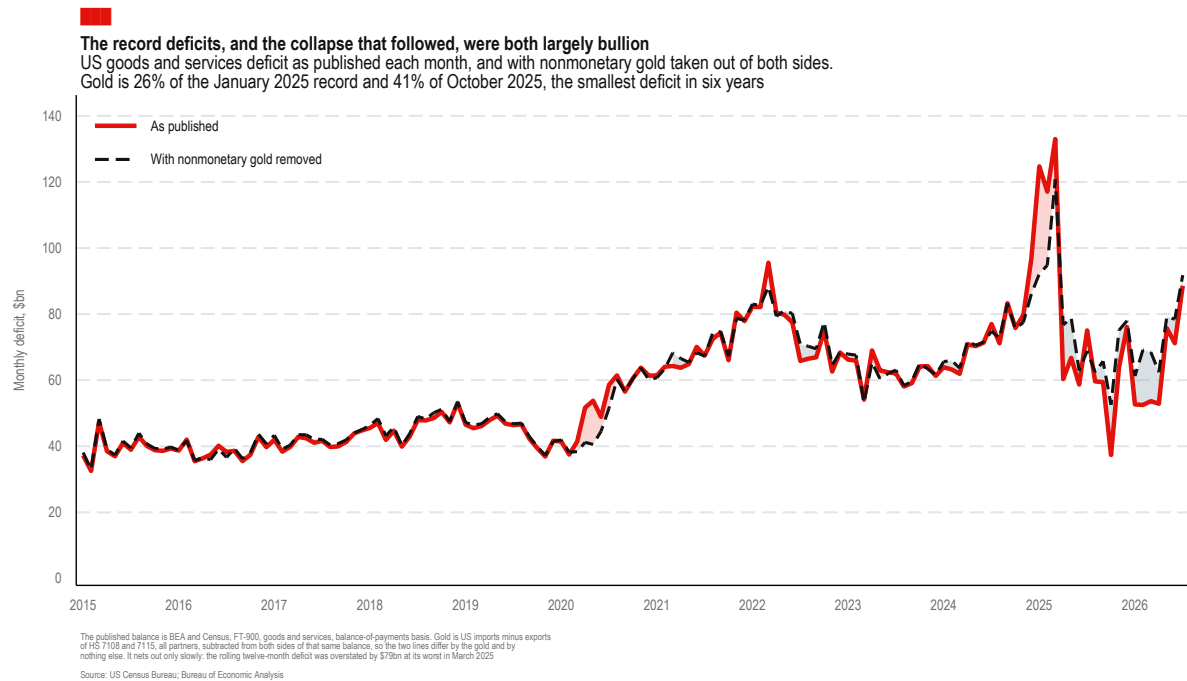
The natural reassurance is that a round trip nets out over a year and only the monthly print is affected. That reassurance is wrong, and it is worth stating plainly because it is the first thing most readers will reach for. At its March 2025 peak the rolling twelve-month deficit was overstated by **\$79.5 billion**, 7.4% of the trailing-year total, because a full year of data then contained the entire surge and none of the reversal. Eleven months after the metal began moving, the twelve-month figure was still overstated by \$16 billion. The distortion is slower than the annual frequency at which most people read trade data.

Second, through the deflator

Everything above is nominal. The release also publishes a real series — Exhibit 10, real goods trade by end-use category in chained 2017 dollars — and gold reaches that one differently.

Real trade is nominal trade divided by a price index, and the index is a modified Laspeyres: fixed base-period quantities. BLS does sample nonmonetary gold, so the problem is not coverage. It is weight. A fixed-weight index carries gold at roughly its base-period share of trade, and that share has not held still: it averaged 0.55% of U.S. goods imports over 2015–19, 1.5% from 2020, and reached **10.5% in January 2025**.

A component weighted near half a percent and actually accounting for ten cannot be tracked. The arithmetic is the standard index-number term — the published inflation rate misses $(s_t - w_{\text{base}})$ times the gap between gold and non-gold inflation — and it is zero whenever either factor is zero, which is why none of this mattered before 2020:



4. The U.S. monthly trade deficit as published and with nonmonetary gold removed, 2015–2026

Notes: Gold is U.S. imports minus exports of HS 7108 and 7115, all partners, subtracted from both sides of the published balance. Shading is red where gold widened the deficit and grey where it narrowed it; the deficit is plotted as a positive number, so up is wider.

Sources: U.S. Census Bureau; Bureau of Economic Analysis.

mean monthly correction, to 2019	0.001%
mean monthly correction, 2024 onward	0.036%
cumulative since January 2020	+2.4 points

That is about an eighth of all measured import price inflation since 2020. February 2025 is the clearest single month: gold was 8.4% of goods imports and rose 6.6%, so at its actual share it should have added 0.55 points on its own. The whole index moved 0.21%.

An understated deflator means an overstated volume, and the overstatement is not confined to gold. The error enters through one end-use category — industrial supplies and materials, where Census books the bars — and is carried into the real goods total. The evidence that the category became unusable is that a practitioner stopped using it: the Atlanta Fed discards the advance report’s industrial supplies figure outright, because it “does not disentangle either the Census-basis measure of nonmonetary gold or finished metal shapes from the sub aggregate,” and forecasts around it instead.⁴

The national accounts already make this adjustment

There is a straightforward precedent for what we are suggesting, and it is already in force one department away.

The Bureau of Economic Analysis does not allow nonmonetary gold into the national accounts. Its stated treatment is to *remove* exports and imports of nonmonetary gold from the international transactions data and replace them with an adjustment computed as domestic production less industrial use.⁵ The reasoning is exactly the reasoning of this letter: a bar bought as a store of value is a *valuable*, not consumption and not investment, so admitting it to gross domestic product would record activity that did not occur. This is long-standing practice, not a response to 2025.

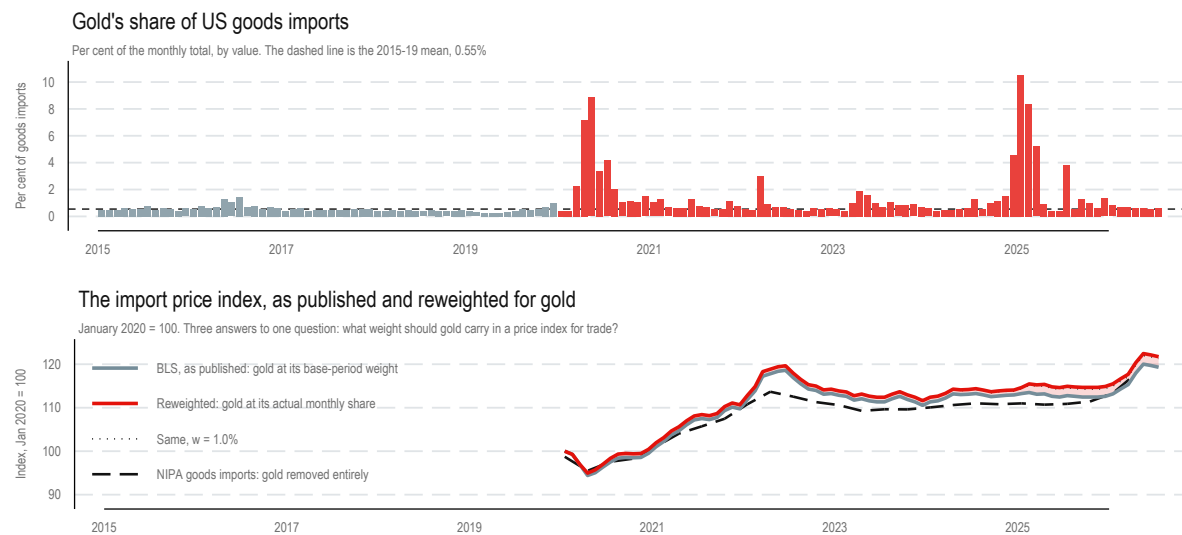
That the rule is live rather than nominal can be seen in what BEA did with a neighbouring metal. In the first quarter of 2025 it identified and removed a surge in imports of silver bars from industrial supplies and materials, because silver bars arrive under a heading the standing gold adjustment does not reach. Gold required no mention because gold comes out as a matter of course.

The trade release applies no such filter. The consequence is that the same metal is excluded from one official statistic and headlined in another, and a reader who moves between the two has no way to see it from the published figures.

The cost of that gap was borne, visibly, by forecasters. A nowcasting model bridges from monthly trade data to a quarterly GDP concept; if the trade series contains gold and the GDP concept excludes it, the bridge is mapping one thing onto another. In early March 2025 the Federal Reserve Bank of Atlanta recalibrated its GDPNow model to strip gold out of the trade aggregates it extrapolates from, ran the standard and gold-adjusted versions side by side for two months, and on 30 April made the gold-adjusted version the standard model.⁶ A respected model had to be rebuilt mid-quarter because a published series measured something its target did not.



A fixed-weight index cannot see a component whose share moved twentyfold
Gold is in the import price index, but at roughly its base-period weight.
Its actual share of US goods imports reached 10.5% in January 2025. Reweighting it at the share it actually had adds 2.4 points to import price inflation since 2020



The correction is the standard index-number term $(s_t - w_{\text{base}}) \times (\text{gold inflation} - \text{non-gold inflation})$, with non-gold inflation backed out of the published index given w_{base} . BLS does not publish the gold weight, so two values bracketing its pre-2020 trade share are shown. This does NOT affect published GDP: BEA removes nonmonetary gold from the national accounts, so they are not deflated through this index. It affects real trade series built from the trade release, and nowcasts bridging from them.

DO NOT READ THE NIPA LINE AS THE GOLD EFFECT. It is quarterly, it excludes gold by construction, and it sits below the published index - but it is also a chain-Fisher index over a different basket, and tested against gold directly the gap does not track it: the correlation between the quarterly NIPA-BLS gap and the predicted gold term is -0.15, and in 2025Q1 the observed gap was seven times the size gold can account for. It is shown because it is what BEA does when it decides gold does not belong in a trade aggregate, not as a measurement of this bias

Source: Bureau of Labor Statistics; US Census Bureau; Bureau of Economic Analysis; LBMA

5. Gold's share of U.S. goods imports, and three answers to what weight it should carry, 2015–2026

Notes: The correction is $(s_t - w_{\text{base}})$ times the gap between gold and non-gold inflation, with non-gold inflation backed out of the published index. The NIPA line excludes gold by construction, but the gap between it and the published index should *not* be read as the gold effect: it is a chain-Fisher index over a different basket, and tested directly the gap does not track gold, correlating -0.15 with the predicted gold term. It is shown because it is what BEA does when it decides gold does not belong in a trade aggregate.

Sources: Bureau of Labor Statistics; U.S. Census Bureau; Bureau of Economic Analysis; LBMA.

Conclusion

The 2025 gold episode is a clean case of a general problem. Trade statistics record border crossings, and a border crossing is an excellent proxy for economic activity right up until the moment it is not. For most goods the proxy holds. For bullion held as a store of value it fails completely: the metal crosses, the statistic registers, and nothing has been produced, consumed or even, in the relevant sense, traded.

The episode was cheap in resources and expensive in information. Roughly \$243 million of freight, insurance, recasting and financing — 0.14% of the value shuttled — bought a distortion that accounted for a quarter of a record monthly deficit, two-fifths of an unusually narrow one, \$79 billion of a trailing-year figure, and something on the order of an eighth of measured import price inflation since 2020.

The remedy does not require a new concept, only the consistent application of an existing one. BEA already computes the nonmonetary gold adjustment for the national accounts, and the underlying series are already collected. Publishing the goods balance on both bases — as reported, and excluding nonmonetary gold — would let users see the wedge at the moment they read the headline rather than reconstructing it afterwards from a classification detail.

The real series needs it more, not less. A determined reader can undo the nominal distortion, because the gold line is published and subtracting it is arithmetic. Nobody outside the agencies can undo the deflator: the weights are not published, so the correction in figure 5 has to be estimated rather than computed. What a reader cannot repair, the publisher has to.

Two features of the data make this more than a presentational nicety. Most of the surge entered the United States under HS 7115, “articles of precious metal,” rather than the heading one would search first; a user pulling HS 7108 alone would have found almost none of the episode.⁷ And the distortion does not wash out at annual frequency, so waiting does not solve it.

None of this is a claim that any published number was wrong. The gold crossed the border, and the balance of payments is designed to record goods crossing borders. The claim is narrower: for this particular good, in these particular quantities, the crossing and the economics came apart, and the statistical system already knows how to tell them apart when it chooses to.

All estimates in this letter are reproducible from public data and one commercial futures feed. The pull scripts, the analysis and the code for each figure are available in the project repository.

Notes

¹Combining Harmonized System headings 7108 and 7115; see note 3.

²COMEX gold contracts settle against 100-ounce and kilobar deliverable forms. The London market trades 400-ounce Good Delivery bars. Metal moving between the two therefore passes through a Swiss refinery, which is why a bilateral U.S.–Switzerland series picks up so much of this.

³The freight, recasting and handling components are built up separately in figure 2’s hurdle, where they total \$0.75 an ounce: \$0.35 of published COMEX depository delivery-out charge — \$35.00 per 100-ounce contract, with delivery in free — plus \$0.20 of air freight and \$0.20 of recasting at the institutional rates

reported during the episode. The wider range quoted here adds insurance and transit financing, which scale with the gold price and therefore roughly doubled over this sample. We did not obtain carrier quotes; a single one would be the cheapest available improvement to both figures. Retail parcel rates of \$300–500 a kilogram circulate widely and are not used: they price a one-kilo consignment to a private buyer rather than a tonne moving between bullion banks, and are two orders of magnitude too high.

⁴Two caveats belong with the 2.4 points. BLS does not publish the gold weight inside the all-commodities index, so w_{base} is assumed; figure 5 draws two values bracketing gold's pre-2020 trade share and the answer moves little between them. And the estimate is made on the aggregate BLS index rather than on Exhibit 10 itself, which is Fisher-chained over end-use categories, so it indicates the size of the effect rather than measuring that series directly.

⁵See BEA, “How are exports and imports of nonmonetary gold treated in BEA's National Economic Accounts?”, and the NIPA Handbook, chapter 8, with the reconciliation in table 4.3C.

⁶Federal Reserve Bank of Atlanta, “Adjustments for Gold Imports and Exports in Standard GDPNow Model.” Over the two months both models ran, the gap between them was about two percentage points of annualized growth.

⁷BEA reclassifies line 7115900530 — rectangular shapes, 99.5% or more gold, not otherwise decorated — as nonmonetary gold on a balance-of-payments basis, while the Census classification treats it as a finished metal shape. The two agencies disagree about what this metal is, which is why a Census-basis pull must add 7115 back by hand.